Saiteja Venkateshwa Rao Dasari

Robotics Engineer | Controls Engineer | Autonomy Engineer

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Summary

M.S. Robotics candidate with robotics and UAV integration experience and projects across ROS2 autonomy, UAV GNC, fault-tolerant control and simulation-to-hardware testing. Strong in Python, C/C++, MATLAB/Simulink, Linux, controls, path planning, and real-time robotic systems.

Technical Skills

Programming/Tools: Python, C/C++, MATLAB/Simulink, Bash, Linux, Git, SolidWorks.
Robotics/Simulation: ROS1/ROS2, Gazebo, MuJoCo, CrazySim SITL, RViz, tf2, URDF/xacro, ArduPilot, Mission Planner, cflib.
Controls/GNC/Test: PID, closed-loop control, state estimation, sensor fusion, FDI, control allocation, controller reconfiguration, SITL, flight-log analysis.
Autonomy/Perception: A*, Hungarian algorithm, path planning, costmaps, occupancy grids, collision avoidance, OpenCV, YOLOv8, ArUco, pose estimation.

Experience

Honeywell Aerospace — Aug 2025 – Apr 2026
Research Externship Role / Technical Lead, SkySpeak AI – ASU Collaborative Research — *Tempe, AZ*

- Directed a 3-engineer team and partnered with Honeywell/ASU stakeholders to deliver an ATC/pilot simulator with LLM agents, structured guardrails, and a 300 ms LiveKit/Deepgram/ElevenLabs voice pipeline.
- Engineered Python voice analytics with YIN pitch detection, VAD, z-score calibration, and abstention scoring (< 0.60); launched a CBTA dashboard for 20+ stakeholder reviews and seed-grant support.

Arizona State University — Jan 2025 – May 2026
Graduate Teaching Assistant, MATLAB Programming and Instrumentation & Controls Lab — *Tempe, AZ*

- Mentored 90+ students across MATLAB, instrumentation, and controls labs; troubleshot code/hardware setups, led reviews, and reinforced PID/control-system concepts.

Marut Drones (IIIT-Hyderabad) — Aug 2023 – May 2024
Production Engineering Intern – UAV Systems Integration — *Hyderabad, India*

- Assembled and verified UAV subsystems including ESCs, flight controllers, GPS, avionics, and propulsion hardware, raising production efficiency by 15% across multirotor platforms.
- Tuned ArduPilot/Mission Planner PID parameters from flight logs across 3 UAV variants; root-caused stability issues and increased flight stability by ∼10%.

Projects

Fault-Tolerant Quadrotor Control – Crazyflie 2.1, MuJoCo, CrazySim SITL — Jan 2026 – May 2026

- Derived and validated closed-form 3-motor allocation formulas for all 4 single-motor failure cases, reaching 0° attitude error across 24 simulated test conditions; root-caused attitude loss to motor saturation and added a thrust clamp that cut max roll from 145° to 7.2° on hardware.
- Built a MuJoCo rigid-body simulation modeling gyroscopic coupling and validated sim-to-real transfer across PID, INDI, and gyroscopic-compensation controllers, testing 9 allocation methods on Crazyflie hardware.

Multi-Robot Task Allocation & Navigation – ROS2, Gazebo, Python/C++ — Dec 2025

- Developed a centralized ROS2/Gazebo autonomy stack for 5 TurtleBots with from-scratch Hungarian assignment, a custom A* planner (ROS2 service), and an inflation costmap; attained 99% mission success.
- Built a path cache keyed on start pose and task ID to align allocated and executed routes, cutting redundant A* re-planning.

Autonomous Drone Landing on Moving Platform – MATLAB/Simulink, PID Control — May 2025

- Built a Parrot Mambo autonomous landing system using HSV color segmentation, centroid tracking, velocity-sync logic, and staged PID descent to align with a moving 11-by-11 in platform; achieved < 5 cm lateral error, ∼4 s landing time, and $> 90\%$ landing success across varied platform speeds.

Crop Weed Detection – ROSMaster X3, ROS, YOLOv8, OpenCV — May 2025

- Deployed real-time YOLOv8 crop/weed detection on ROSMaster X3 using 3,000 samples; connected ROS inference/visualization nodes and reached 97% accuracy at 12 FPS.

Maze Solving & Path Planning – myCobot Pro 600, OpenCV, A*, IK — Dec 2024

- Built an OpenCV vision pipeline into a 34x34 occupancy grid, then computed A* paths and inverse kinematics (MATLAB RigidBodyTree) to execute waypoint trajectories on a physical 6-DOF myCobot Pro 600 arm.

Education

Arizona State University — Aug 2024 – May 2026
M.S. Robotics and Autonomous Systems, Systems Engineering; GPA: 3.93/4.0 — *Tempe, AZ*
Coursework: Aerial Robotics, Mechatronics, Multi-Robot Systems, Controls and Systems.

Chaitanya Bharathi Institute of Technology — Aug 2020 – May 2024
B.E. Electrical and Electronics Engineering — *Hyderabad, India*

Certifications & Publication

Universal Robots e-Series Core/Pro | MathWorks AI | Hugging Face Reinforcement Learning | Udemy ROS/ROS2 Gazebo | SCADA-Based Substation Control Panel and Comprehensive Operations